EXPLORATORY DATA ANALYSIS

IMDb Movie Runtimes & Audience Ratings

Evaluating the relationship between movie length, vote confidence, and audience ratings.

SLIDE 1: OVERVIEW

Dataset Scope & Baseline Metrics

Analyzed Scope

65,720 rated movies evaluated from 2010 to 2019.

Filtered Data

Removed 31,739 missing runtimes and 5,408 missing genres.

Core Baselines

Median runtime of 91 minutes with a 6.5/10 median rating.

SLIDE 2: BUSINESS QUESTION

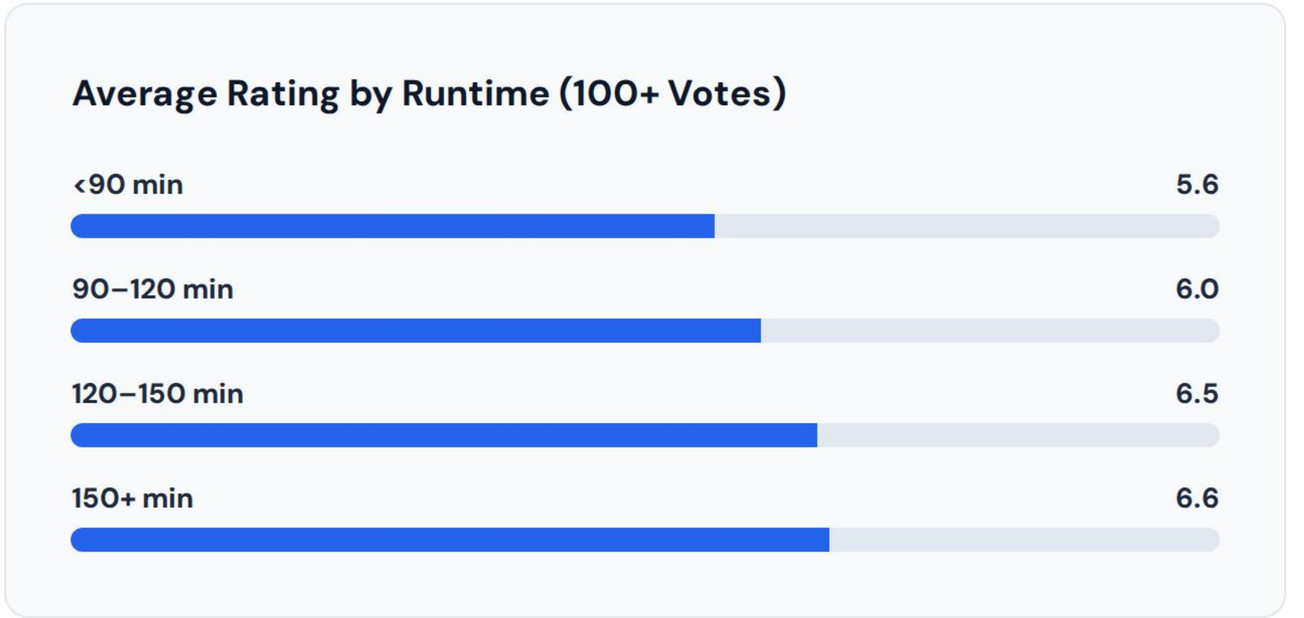
Do Longer Movies Earn Higher Ratings?

Strategic Question

Should studio budgets favor longer runtime productions?

Key Takeaway

Filtering for 100+ votes reveals higher audience ratings for longer runtimes.



SLIDE 3: DATA PREPARATION

Key Data Cleaning Steps

Null & Date Filtering

Dropped missing runtime and genre rows; excluded releases after 2019.

Outlier Treatment

Handled runtime extremes (max 51,420 min; 290 short films under 40 min).

Data Integrity

Enforced a 100+ vote threshold and utilized unique `movie_id` primary keys.

Thank You

Questions & Strategic Discussion